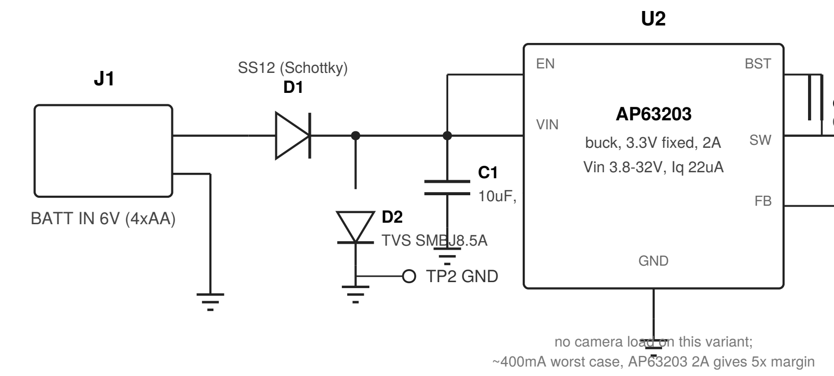
ESP32-C6-MINI-1-N8 Doorlock Baseboard — DLBASIC V1.0

Minimal variant: J1 + J2 only, no camera/I2C • target pocket 42 x 23 x 10mm deep • Sheet 1 of 2 (electrical)

OPEN: J2 connector type/pitch and SRDY/MRDY wake convention (level vs. pulse) not yet confirmed for this lock model — see spec §2.

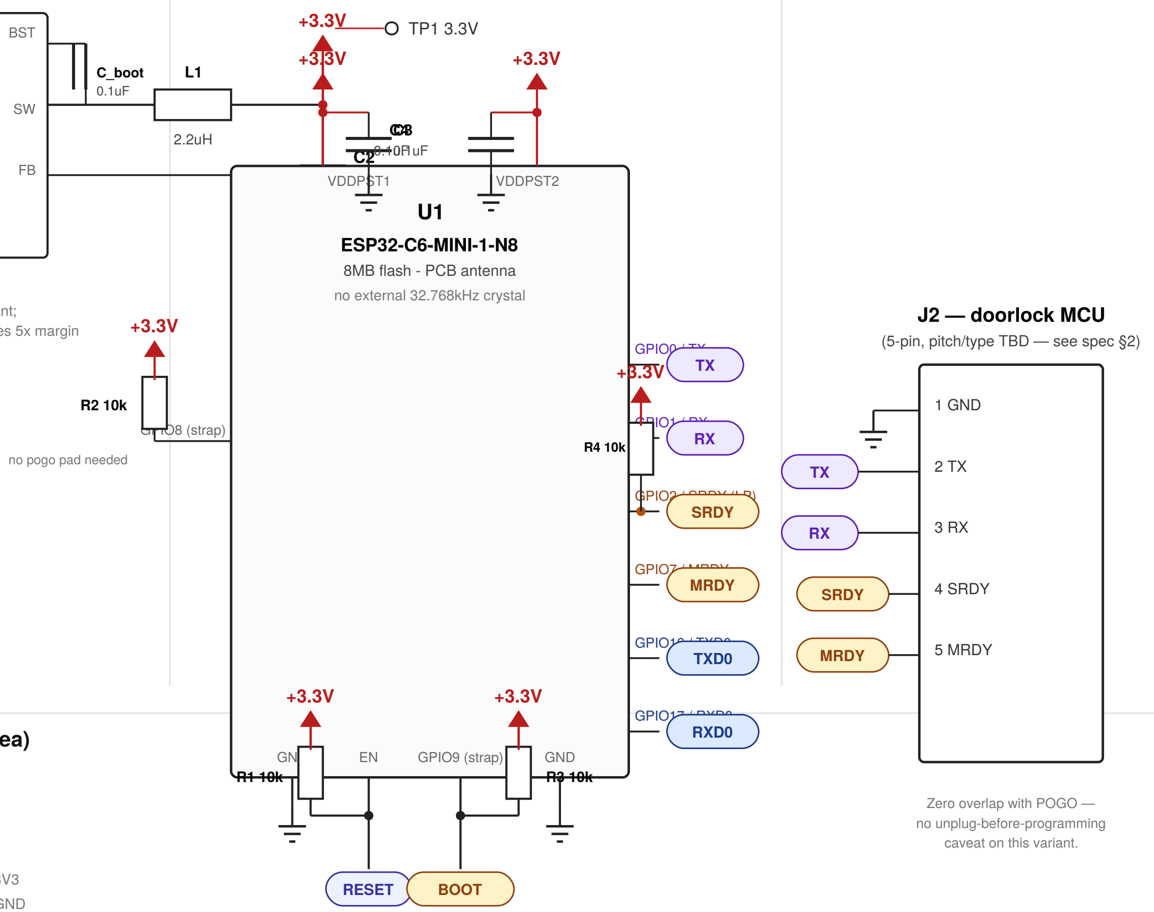
Programming fully separated from J2: POGO uses GPIO16/17 (true ROM UART), J2 uses GPIO0/1 — zero pin overlap, no "unplug before flashing" caveat.

POWER SUPPLY (buck)



CORE MODULE

J2 — MCU INTERFACE (5p)



NOTES

Design basis

DLBASIC V1.0 — minimal variant of the Premium doorlock+camera baseboard
Target pocket: 42 x 23 x 10mm deep

2.1 J2 connector — open

Pitch/type not yet confirmed for this lock model. Shown generic pending spec.

2.2 Wake convention — open

Level-held vs. pulse for SRDY/MRDY not yet confirmed. ESP32-C6 EXT1 deep-sleep wake supports ANY_HIGH/ANY_LOW natively — either convention is a firmware config, not a hardware change.

2.3 Programming fully isolated

GPIO16/17 (true ROM UART) go only to POGO. J2 uses GPIO0/1/2/7. No shared pins between production and programming.

Removed vs. Premium

J3 (I2C+spare), J5 (camera SPI) — not present on this variant

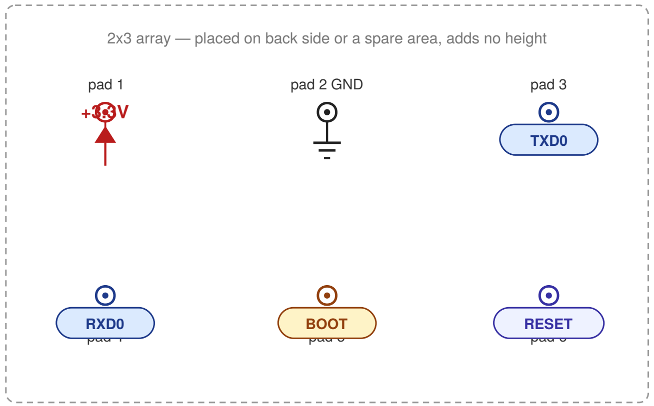
Symbol key

ground

+3.3V
+3.3V tap

net label (same color+name = same net)

POGO — PROGRAMMING (6-pad, 2x3, back side or spare area)

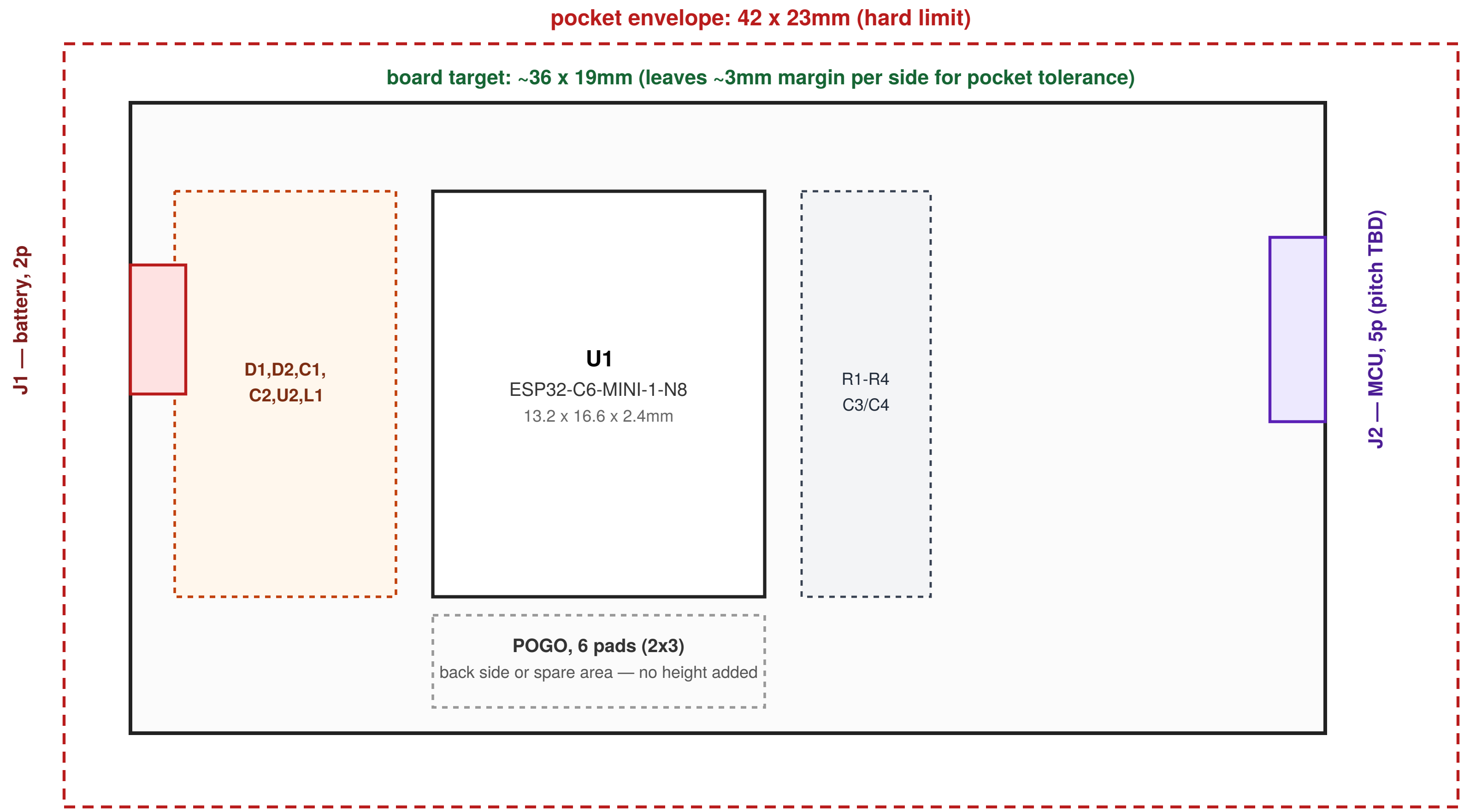


pad 1 = 3V3
pad 2 = GND
pad 3 = TXD0 (GPIO16)

pad 4 = RXD0 (GPIO17)
pad 5 = BOOT (GPIO9)
pad 6 = RESET (EN)

Mechanical Layout — DLBASIC V1.0 (42 x 23 x 10mm deep pocket)

2D top-down placement, illustrative • target board well inside the pocket envelope for tolerance margin • Sheet 2 of 2



Dashed red = hard pocket limit (42 x 23mm). Board and all components must clear this with margin for insertion tolerance.

10mm depth budget — needs explicit component-height check

42 x 23mm is comfortable for this pin count (7 populated pins + 6 bare pads) — board area is not the constraint here. Depth is: the real risk is stack height — PCB (~1.6mm) + tallest component + connector mating height + any standoff, all within 10mm total. Candidates to check once parts are chosen: J1/J2 connector body height (especially if THT with a mating plug on top), L1 inductor height, and battery connector height if it protrudes above the PCB plane.

Action: get real component height specs before finalizing part selection — this is the binding constraint on this variant, not board area.

Placement summary

J1 (battery, 2-pin): left edge
J2 (doorlock MCU, 5-pin, connector type/pitch TBD): right edge
POGO (6-pad, 2x3, programming): interior, back side or spare area — zero height impact

No J3/J5 on this variant. Mounting method not yet specified — confirm with lock manufacturer.

ESP32-C6-MINI-1-N8 Doorlock Baseboard — Engineering Specification

DLBASIC V1.0 — minimal variant, 42 x 23 x 10mm pocket · companion sheets: schematic (p.3) + mechanical layout (p.4)

Two items still open before this can be finalized: J2's connector type/pitch, and the SRDY/MRDY wake convention (level-held vs. pulse) for this specific lock model's MCU firmware. Both are documented as open in the sections below rather than guessed at. Everything else — including full separation of programming from the production interface — is locked in.

1. Overview

DLBASIC V1.0 is a minimal, connector-light variant of the Premium doorlock+camera baseboard, built for a lock model with a tight 42 x 23 x 10mm-deep mounting pocket. It shares the same power supply architecture (6V/4xAA into an AP63203 buck) and core module as Premium, but carries only the two connectors this application needs: battery input and a 5-pin doorlock MCU interface. Camera (J5) and I2C/spare-GPIO (J3) are not present on this variant.

- **Core:** ESP32-C6-MINI-1-N8, same power supply and strapping design as Premium
- **J1 (2-pin):** 6V (4xAA) battery input, unchanged from Premium
- **J2 (5-pin):** GND, TX, RX, SRDY, MRDY — pure MCU interface, connector type/pitch TBD
- **POGO (6-pad, 2x3):** factory/depot programming, fully isolated from J2 — zero shared pins

2. Design notes

2.1 J2 connector — still open

Type and pitch not yet confirmed for this lock model. This project now has three different connector conventions across its variants (2.54mm pogo/pad-style, 2.00mm THT 2x3, and whatever this new lock model expects) — rather than guess, J2 is documented here as a generic 5-position footprint pending confirmation. This does not block electrical design (GPIO assignment is settled) but does block final PCB footprint placement.

2.2 SRDY / MRDY wake convention — still open

Level-held (e.g. "MCU pulls low until acknowledged") vs. pulse (e.g. "300ms high pulse") not yet confirmed for this lock model's MCU firmware. Confirmed via ESP-IDF documentation: ESP32-C6's deep-sleep GPIO wake (EXT1) natively supports both ANY_HIGH and ANY_LOW level configuration — either convention is a firmware config choice, not a hardware change, and a pulse of a few hundred milliseconds or more will reliably register against a level-configured wake input. No hardware risk either way; only firmware needs the confirmed answer.

2.3 Programming fully isolated from J2 (locked in)

Unlike Premium (where J2 doubles as the programming interface and requires physically unplugging the doorlock MCU before flashing), DLBASIC dedicates GPIO16/GPIO17 (the true ROM bootloader UART) exclusively to the 6-pad POGO cluster, and J2's UART runs on GPIO0/GPIO1 instead. There is no pin overlap and no unplug-before-programming procedure needed on this variant.

2.4 10mm pocket depth — needs component-height confirmation

Board area is not the constraint on this variant (7 populated connector pins + 6 bare pads fit 42 x 23mm comfortably, target board ~36 x 19mm). Depth is the real risk: PCB thickness + tallest component + any connector mating height must fit within 10mm total. This needs real part heights for J1, J2, and L1 before the variant can be considered mechanically closed — see sheet 2.

3. Pin mapping

U1 pin	Physical pin	Function	Connects to	Notes
EN (CHIP_PU)	8	Reset	R1 pull-up 10k; POGO pad 6	unchanged from Premium
GPIO8	22	Strap — ROM print control	R2 pull-up 10k only	no POGO/J2 pin needed
GPIO9	23	Strap — boot mode select	R3 pull-up 10k; POGO pad 5	must be forced LOW for download
GPIO0	12	UART TX (J2)	J2 pin2	free to reuse — no bootloader conflict on this variant
GPIO1	13	UART RX (J2)	J2 pin3	free to reuse — no bootloader conflict on this variant
GPIO2	5	SRDY	J2 pin4; R4 pull-up 10k	LP_GPIO — deep-sleep wake-capable
GPIO7	16	MRDY	J2 pin5	plain GPIO, no wake requirement
GPIO16	31	TXD0 (programming)	POGO pad 3	true ROM download TX
GPIO17	30	RXD0 (programming)	POGO pad 4	true ROM download RX

Physical pin numbers derived directly from the ESP32-C6-MINI-1 module datasheet pin table.

4. Net list

Net	Nodes
VBAT_IN	J1.1, D1 anode
VIN_PROT	D1 cathode, D2, C1+, U2.VIN, U2.EN
+3.3V	U2.VOUT(FB node), C2+, U1.VDDPST1, U1.VDDPST2, C3+, C4+, R1-R4 tops, POGO pad1, TP1
GND	J1.2, D2 cathode, C1-, U2.GND, C2-, U1.GND (all), C3-, C4-, J2.1, POGO pad2, TP2
SW node	U2.SW, L1 pin1, C_boot (to BST)
EN	U1.EN, R1 bottom, POGO pad6

Net	Nodes
GPIO8	U1.GPIO8, R2 bottom
GPIO9 / BOOT	U1.GPIO9, R3 bottom, POGO pad5
TX	U1.GPIO0, J2.2
RX	U1.GPIO1, J2.3
SRDY	U1.GPIO2, J2.4, R4 bottom
MRDY	U1.GPIO7, J2.5
TXD0	U1.GPIO16, POGO pad3
RXD0	U1.GPIO17, POGO pad4

5. Bill of materials

Ref	Qty	Description	Value / part	Package	Example MPN	Notes
U1	1	Wi-Fi 6 / BLE5 / 802.15.4 module	ESP32-C6-MINI-1-N8	SMD module, castellated	ESP32-C6-MINI-1-N8	Unchanged from Premium
U2	1	Synchronous buck converter, 3.3V	AP63203	SOT23-6	AP63203WU-7	2A, Vin 3.8-32V, Iq 22uA
L1	1	Buck inductor	2.2uH	SMD	—	Confirm height vs. 10mm depth budget — see §2.4
C_boot	1	Bootstrap capacitor	0.1uF	0402/0603	—	Required between BST and SW
D1	1	Schottky diode, reverse-polarity protection	SS12 (1A, 20V)	SMA / DO-214AC	SS12-E3/61T	Unchanged
D2	1	Bidirectional TVS, ESD/surge protection	SMBJ8.5A	SMB / DO-214AA	SMBJ8.5A-13-F	Sized for 6V (4xAA) input
C1	1	Bulk input capacitor	10µF X5R 16V	0805	—	
C2	1	Bulk output capacitor	22µF X5R 16V	0805	—	
C3, C4	2	Decoupling, near U1 power pins	0.1µF X7R 16V	0402	—	Unchanged
R1	1	EN pull-up	10kΩ, 1%	0402	—	
R2	1	GPIO8 pull-up	10kΩ, 1%	0402	—	
R3	1	GPIO9 pull-up	10kΩ, 1%	0402	—	

Ref	Qty	Description	Value / part	Package	Example MPN	Notes
R4	1	SRDY pull-up	10kΩ, 1%	0402	—	
J1	1	Battery connector, 2-pin	TBD, matches Premium unless told otherwise	THT, 2-pin	—	Confirm height vs. 10mm depth budget
J2	1	Doorlock MCU connector, 5-pin	TBD — see §2.1	TBD	—	Type/pitch not yet confirmed for this lock model
—	—	J3 (I2C+spare), J5 (camera SPI)	NOT PRESENT	—	—	This variant only — see Premium spec for camera-capable design

6. Power budget (sanity check)

- No camera load on this variant — worst case is module active Wi-Fi TX alone, roughly 300-450mA class. AP63203's 2A rating gives well over 4x margin.
- Same buck efficiency and quiescent-current advantages as Premium apply here — see Premium spec §7 for the full analysis.

7. Programming & bring-up procedure

1. Factory/depot: fixture contacts the 6 POGO pads directly. Pulls pad5 (BOOT/GPIO9) low, pulses pad6 (RESET/EN), flashes over pads 3/4 (TXD0/RXD0 = GPIO16/GPIO17).
2. J2 is never involved in programming — no need to disconnect the doorlock MCU at any point, unlike Premium.
3. After programming, connect J2 to the doorlock MCU and verify the UART + SRDY/MRDY handshake once the wake convention (§2.2) is confirmed.

8. Open items before this variant is fab-ready

- Confirm J2 connector type and pitch for this lock model — see §2.1. Blocks final footprint.
- Confirm SRDY/MRDY wake convention (level vs. pulse, and exact timing if pulse) for this lock model's MCU firmware — see §2.2.
- Get real component height specs for J1, J2, and L1 against the 10mm pocket depth — see §2.4. This is the binding mechanical constraint on this variant.
- Confirm mounting method (adhesive, screw bosses, friction-fit, etc.) with the lock manufacturer — not yet specified.